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EE9-AML1-01

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

EEE/EIE PART III/IV: MEng, BEng and ACGI

MACHINE LEARNING

Friday, 6 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer THREE questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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MACHINE LEARNING EXAM PAPER 2024

PLEASE ANSWER 3 OUT OF 4 QUESTIONS.

1. Linear Classification, Closed form solution, Non-linear Feature Transformation, VC Dimension & Perceptron Learning Algorithm

a) You are given a dataset with 600 samples of movies, each rated on a scale from 1 to 5 (see Table 1 for sample data entries). The dataset contains the following features for each movie:

- Movie Age: Age of the movie in years (calculated from January 2024).
- Movie Length: Duration of the movie in hours.
- Movie Budget: Movie budget in millions of pounds.
- Box Office: Box office earnings in millions of pounds.

Movie Age	Movie Length	Movie Budget	Box Office	Rating
5	2	10	25	4
2	1.5	7.5	15	3
15	2.5	5	10	2
1	1.8	15	30	5
⋮	⋮	⋮	⋮	⋮
2	1.5	7.5	15	2

Table 1 Sample of movie rating data.

Your task is to predict the movie rating using historical data including features and labels. Answer the following:

i) Is this a supervised or unsupervised machine learning problem? Explain why. **[2 marks]**

ii) How can we tackle this problem either as regression or classification? Justify your answer. **[2 marks]**

iii) Explain when can stochastic noise occur. Give an example from your dataset. **[2 marks]**

iv) Suppose you decide to simplify the prediction by classifying movies as "positive" if their rating is 3 or higher and "negative" if their rating is below 3. Assume you are using a Perceptron Learning Algorithm

(PLA) with the same feature space to solve this binary classification problem, answer the following:

What will be the VC dimension of the hypothesis class when applying the perceptron model given? **[3 marks]**

Using the rule of thumb learned in the class, comment whether the data samples are considered sufficient to train the perceptron. **[3 marks]**

Describe the main challenge you may encounter in finding the best hypothesis using the PLA and suggest two solutions to reduce it (assuming that you have to use PLA). **[3 marks]**

b) Consider the truth table of AND gate given in Table 2.

$\mathbf{x_1}$	$\mathbf{x_2}$	**y**
-1	-1	-1
-1	+1	-1
+1	-1	-1
+1	+1	+1

Table 2 Truth table for AND gate.

Suppose you want to use a perceptron to build an AND gate classifier. Apply the perceptron learning algorithm (PLA) you learned in the class to find the parameters of this perceptron. Then prove why the resulting hypothesis by PLA and the perceptron given by $h \rightarrow [w_0, w_1, w_2] = [-1.5, 1, 1]$ can be both accepted as a solution. **[15 marks]**

c) Consider a classifier called positive-ray that is given by the following hypothesis:

$$h(x) = \text{sign}(x - a) \text{ s.t. } x \in \mathbb{R}^1$$

Point a acts as decision boundary in 1-D space, where any data point can be only classified within the positive class if it lies exactly on a or on its right hand side (see Figure 1.1).

Using the bound $m_{\mathscr{H}} = \sum_{i=0}^{k-1} \binom{N}{i}$, where k is the break-point, derive the polynomial that represents the maximum number of dichotomies that can be obtained using this classifier in terms of N (the number of samples). Then

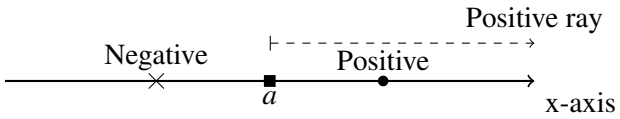


Figure 1.1 Positive-ray classifier.

using the polynomial obtained find the maximum number of dichotomies when $N = 100$. **[15 marks]**

d) Consider the data samples along with their ground truth values y shown in Table 3.

x_1	x_2	y
2	2	-1
4	2	-1
4	3	-1
1	2	+1
3	1	+1
2	3	+1

Table 3 Data sample.

i) Plot the data on a Cartesian plane and prove that they are non-linearly separable and cannot be classified with zero error using a linear classifier (e.g., a perceptron). **[10 marks]**

ii) Apply the regression for classification approach by using closed form solution to find the weight vector. State whether you would consider the hypothesis you found as good solution by estimating the correctly classified and misclassified samples and commenting on your findings. **[20 marks]**

iii) Apply the non-linear transformations below:

$$\Phi(\mathbf{x}) = \begin{cases} \left(x_1 x_2,\ \frac{x_1^2}{4}\right) & \text{if } (x_1 x_2 \bmod 4) = 0 \\ (x_1,\ x_2) & \text{if } (x_1 x_2 \bmod 4) \neq 0 \end{cases}$$

Note: mod is the operation that returns the remainder of the division.

and answer the following:

- State whether this transformation can make the data linearly separa-

ble. Then mention if this transformation retained the original feature space or not. **[15 marks]**

- Suppose that there is a perceptron to classify the newly transformed points given by its weights $[w_0, w_1, w_2, ..., w_n]$ as follows $h \rightarrow \mathbf{w} = [12, -3, -1]$. Estimate the confusion matrix by applying the hypothesis (reporting true positives, true negatives, false positives and false negatives) and total accuracy. **[10 marks]**

2. Linear regression, Neural Networks, Backpropagation, Gradient Descent, Regularization, Validation, and Bias-Variance Analysis

a) Consider a sample dataset of Iris flowers with features "sepal length" and "petal length", and target classes "Setosa" (labeled as 1) and "Virginica" (labeled as 0). Table 4 contains the training samples extracted from this dataset.

Sepal Length in cm ($\mathbf{x_1}$)	Petal Length in cm ($\mathbf{x_2}$)	Flower Type ($\mathbf{y}$)
5	1.5	1
6	5	0

Table 4 Training samples.

i) Train a logistic regression classifier to classify the training data using batch gradient descent over two steps with a learning rate of 0.1. Assume all the weights are initialised with zero at the beginning and the binary cross entropy function that you will use is given by:

$$\text{BCE} = \frac{1}{N}\sum_{i=1}^{N}[-y_i \log(\hat{y}_i) - (1-y_i)\log(1-\hat{y}_i)]$$

In your solution, state clearly the gradient, update rules used and the updated weights after each step. **[20 marks]**

ii) Suppose that a neural network has been trained using the training data found in Table 4, following the architecture and weights as illustrated in Figure 2.1. The trained network is structured with an input layer comprising two inputs, represented as (x_1) and (x_2), and bias. The hidden layer consists of bias and three neurons denoted as (a_1), (a_2), and (a_3), all using tanh activation functions. The output layer features a single neuron marked as (b), which utilises a sigmoid activation function. Apply feedforward operation to get the predicted values for the training data, then estimate the training error using binary cross entropy error (BCE), and mention if overfitting is likely to occurr based on your results and the training scenario. **[20 marks]**

iii) Apply backpropagation algorithm with stochastic gradient descent to calculate one update of weight $w_{23}^{(1)}$ of the network in Figure 2.1 using the data point $x = [x_1, x_2] = [6, 5]$ with its label $y = 0$. Assume that we are using the same weights in the figure with the same activation functions as in the previous part and a learning rate of 0.1. For your

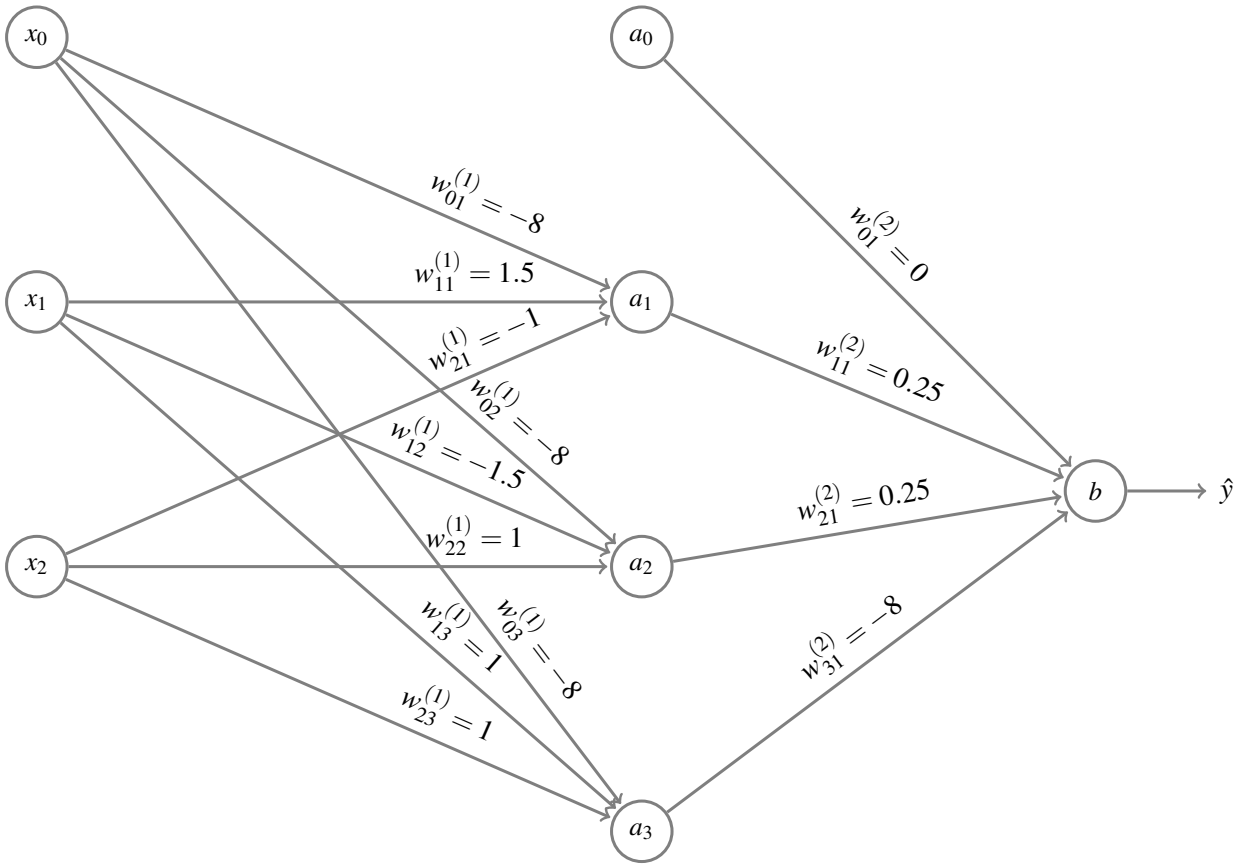


Figure 2.1 Trained neural network (x_0 *and* a_0 *are both equal to 1 and they are used to accomdate the bias term*).

analysis, consider using BCE. Show in your solution all the equations and steps (include all the δ's used in the back-propagation formula and update rule) and discuss the impact of the update and whether the difference is significant (with reasons why it is significant or not). **[15 marks]**

b) Consider the data samples along with their ground truth values y shown in Table 5.

x	y
1	2
3	5
5	3

Table 5 Data sample.

i) Apply ridge regression with $\lambda = 0.1$ to estimate $\mathbf{W}_{reg}$. **[10 marks]**

ii) Using linear regression with leave-one-out cross validation. Estimate the total cross validation error such that each time the model is trained on $n-1$ samples will generate a training error of zero (use MSE function as your error function). **[20 marks]**

c) Assume that two neural networks were trained on the same dataset, the performance of both networks were evaluated based on the validation and training error values as shown in Figures 2.2 and 2.3.

i) Which model would you choose and why? **[5 marks]**

ii) Mention the feasibility of applying early stopping for the training in both networks, mention when and why? **[5 marks]**

iii) Which model is more likely that uses regularization? **[5 marks]**

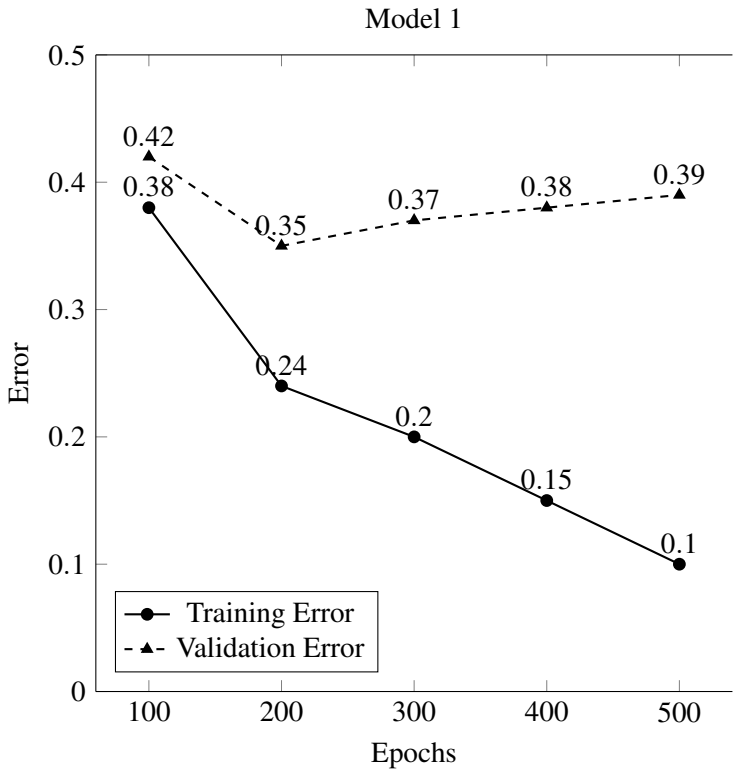


Figure 2.2 Neural network training on model 1.

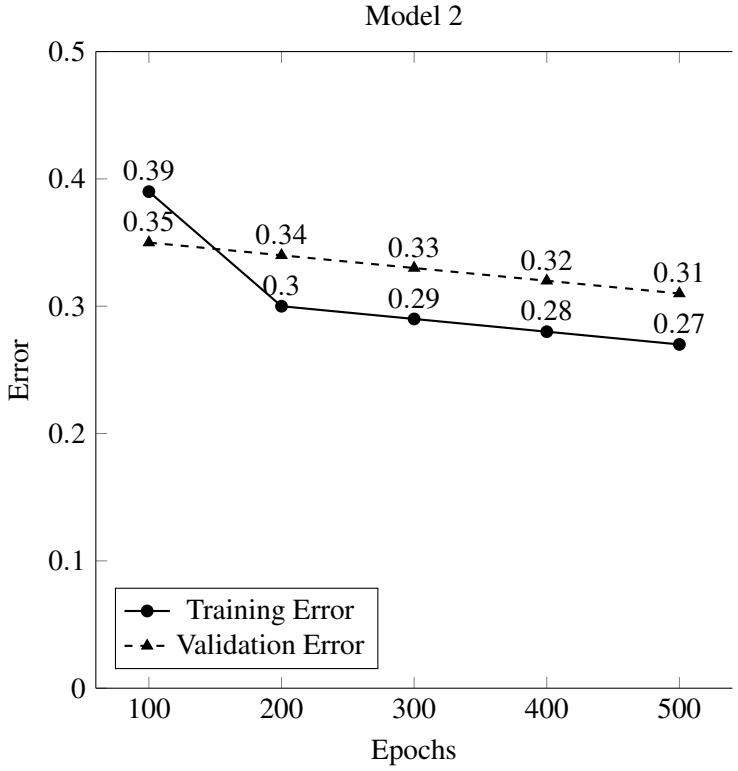


Figure 2.3 Neural network training on model 2.

3. Support vector machines, kernel methods

a) Consider the following dataset:

Positive examples: [-4, -3], [-3, -2], [-1, -2], [-2, 0]

Negative examples: [1, 0], [2, 3], [3, 1], [2, 0]

i) Mark these points on a plot, and identify the maximum margin classifier. Write down the equation of the corresponding hyperplane in the form $\mathrm{w}^{\mathrm{T}}\mathrm{x}+\mathrm{b}=0$, where $\|\mathrm{w}\|^2=1$. **[20 marks]**

ii) Identify the support vectors. **[5 marks]**

b) Answer the following questions about support vector machines (SVMs).

i) Consider we have a linearly separable dataset of N samples in a d-dimensional space, and we apply hard-margin SVM. What are the minimum and maximum number of support vectors? **[10 marks]**

ii) Suppose we use the soft-margin SVM formulation, and gradually increase the penalty parameter C. Do you expect the number of support vectors to increase or decrease as C increases? **[10 marks]**

iii) Suppose that we use the formulation for the soft-margin SVM, i.e., with slack variables ξ_i, but the data we have is actually linearly separable. Do we always recover the 'true' max-margin separating hyperplane? **[10 marks]**

c) Let $\mathrm{K}(\mathbf{x}_i,\mathbf{x}_j)=\phi(\mathbf{x}_i)^T\phi(\mathbf{x}_j)$ denote the kernel function for some feature map $\phi(\cdot)$ that maps data points to a higher dimensional feature space. Consider a dataset of 3 points $\mathbf{x}_1,\mathbf{x}_2,\mathbf{x}_3$, where the kernel matrix is given by:

$$\mathrm{K}=\begin{bmatrix}4 & 2 & 1\\ 2 & 2 & 1\\ 1 & 1 & 1\end{bmatrix}.$$

For example, $\mathrm{K}(\mathbf{x}_1,\mathbf{x}_1)=\phi(\mathbf{x}_1)^T\phi(\mathbf{x}_1)=4$ and $\mathrm{K}(\mathbf{x}_1,\mathbf{x}_2)=\phi(\mathbf{x}_1)^T\phi(\mathbf{x}_2)=2$, and so on so forth.

i) Prove that the Mercer conditions are satisfied for K for these three points. **[10 marks]**

ii) Find the squared norm of the mean vector in the feature space $\|\boldsymbol{\mu}_\phi\|^2$, where the mean vector is defined as:

$$\boldsymbol{\mu}_\phi=\frac{1}{3}\sum_{i=1}^{3}\phi(\mathbf{x}_i).$$

[15 marks]

iii) Find the total variance in the feature space, defined as:

$$\sigma_{\phi}^2 = \frac{1}{3}\sum_{i=1}^{3} \|\phi(\mathbf{x}_i) - \boldsymbol{\mu}_{\phi}\|^2$$

[20 marks]

4. Principal components analysis, clustering.

a) You are given the following data set:

$$\mathbf{x_1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \mathbf{x_2} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \mathbf{x_3} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

i) Plot the data set. **[3 marks]**

ii) Compute the empirical covariance matrix $\mathbf{S}$. **[6 marks]**

iii) Compute the eigenvectors and corresponding eigenvalues S. What is the unnormalized basis vector $\mathbf{u}_1$ of your 1-dimensional representation? **[10 marks]**

iv) Compute the coefficients z_1, z_2, z_3 of the three samples in the dataset. Compute the lower-dimensional representations $z_1\mathbf{v_1}$, $z_2\mathbf{v_1}$, $z_3\mathbf{v_1}$. **[10 marks]**

v) Why would we want to reduce the dimensionality of our data? Can you think of examples where this can be useful? **[5 marks]**

vi) Consider the following design matrix, representing four sample points $X_i \in \mathbb{R}^2$. $X = \{(4,1),(2,3),(5,4),(1,0)\}$. We want to represent the data in only one dimension, so we turn to principal components analysis (PCA). Compute the unit-length principal component directions of X, and state which one the PCA algorithm would choose if you request just one principal component. Please provide an exact answer, without approximation. (You will need to use the square root symbol.) Show your work! **[3 marks]**

vii) The plot in Figure 4.1 depicts the sample points from X. We want a one-dimensional representation of the data, so draw the principal component direction (as a line) and the projections of all four sample points onto the principal direction. Label each projected point with its principal coordinate value (where the origin's principal coordinate is zero). Give the principal coordinate values exactly. **[3 marks]**

Note: Transfer the plot shown in Figure 4.1 into your answer book.

viii) Summarize how you would perform PCA on a dataset $X = \{\mathbf{x_i}\}_{\mathbf{i=1}}^{\mathbf{n}}$ and work with transformed dataset $\{\mathbf{z_i}\}_{\mathbf{i=1}}^{\mathbf{n}}$. **[15 marks]**

b) i) Consider a dataset $\mathscr{D} = \{x_1, x_2, \ldots, x_n\}$ in $\mathbb{R}^d$, and you are tasked with performing hierarchical clustering. Define hierarchical clustering and describe the difference between agglomerative and divisive hierarchical clustering ap-

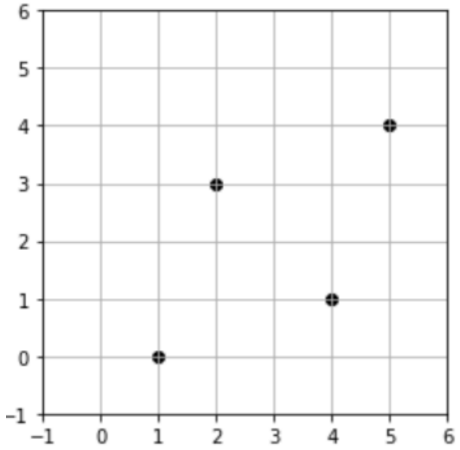


Figure 4.1 Sample points from X.

proaches. **[5 marks]**

ii) Explain the concept of linkage criteria in hierarchical clustering. List and briefly describe at least three different linkage methods. **[5 marks]**

iii) Given a distance matrix D for a dataset with four points, A, B, C, and D, use the single linkage method to perform agglomerative hierarchical clustering. The distance matrix D is:

$$D = \begin{pmatrix} 0 & 2 & 6 & 10 \\ 2 & 0 & 5 & 9 \\ 6 & 5 & 0 & 4 \\ 10 & 9 & 4 & 0 \end{pmatrix}$$

Present the clustering process as a dendrogram and indicate the order of merges. **[10 marks]**

iv) Discuss the strengths and weaknesses of hierarchical clustering compared to k-means clustering. In what scenarios would hierarchical clustering be preferred? **[5 marks]**

c) Suppose you are given the dataset in Table 6 with five points in a 2D space:

i) Use the K-means clustering algorithm with $k = 2$ to partition these points into two clusters. **[1 marks]**

(ii) Let the initial centroids for the two clusters be chosen as:

- Centroid 1: Point A (1, 1)

Point	x	y
A	1	1
B	1	4
C	5	1
D	5	4
E	9	8

Table 6 Sample points in a 2D space.

- Centroid 2: Point E (9, 8)

Assign each point to the nearest centroid. **[1 marks]**

iii) Perform **one iteration** of the K-means algorithm, showing each step clearly:

- Calculate the Euclidean distance of each point to each centroid.
- Assign each point to the nearest centroid.
- Compute the new centroids after this assignment.

[2 marks]

iv) Provide the final cluster assignments for this iteration and the new centroids. **[1 marks]**

c) Let us try and classify data points in 2D Euclidean space. We are given n instances of such points $p_1, p_2, \ldots, p_n$ and corresponding categories $c_1, c_2, \ldots, c_n$ where $c_1, c_2, \ldots, c_n$ take values from all possible class labels. Under the k-nearest neighbours classification scheme, each new element Q is simply categorized by a majority vote among its k nearest neighbors in instance space. The 1-NN is a simple variant of this which divides up the input space for classification purposes into a convex region (see Figure 4.2 for the 1NN decision boundaries under the Euclidean distance measure), each corresponding to a point in the instance set.

i) is it possible to build a decision tree (with decisions at each node of the form is "$x > a$", "$x < b$", "$y > c$", "$y < d$" for any real constants a, b, c, d which classifies exactly according to the 1-NN scheme using Euclidean distance? If so, explain how. If not, explain why not. **[5 marks]**

ii) Now assume that the distance measure is not explicitly specified to you. Instead, you are given a "black box" where you input a set of instances $p_1, p_2, \ldots, p_n$ and a new example Q, and the black box outputs the nearest neighbor of Q, say π and its corresponding class label c_i. Is it possible to construct a k-NN classification algorithm based on this black box alone? If so, how and if not, why

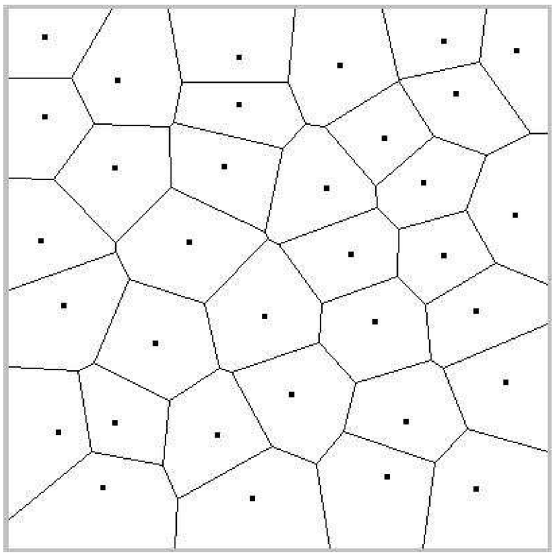


Figure 4.2 Decision boundaries.

not? **[5 marks]**

iii) If the black box returns the j nearest neighbors (and their corresponding class labels) instead of the single most nearest neighbor (assume $j! = k$) , is it possible to construct a k-NN classification algorithm based on the black box ? If so how, and if not why not ? **[5 marks]**